United International University (UIU)

Dept. of Computer Science and Engineering (CSE)

Mid-Term Exam Year: **2025** Semester: **Summer**
Course Code: **CSE 4883** Title: **Digital Image Processing**
Marks: **20** Time: **1 Hour 30 minutes**

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Answer all the questions. All questions are of value indicated on the right-hand margin.

1. Suppose you capture an image of a barcode for a payment. Which is more critical for accurate decoding - sampling rate or quantization levels? Explain briefly. **[2]**

2. Suppose you have charted the performance of your model in the following confusion matrix based on customer feedback classification for a product review platform:

Predicted Review Category	Actual Review Category			
	Positive	Negative	Neutral	Mixed
Positive	800	100	250	50
Negative	50	900	70	120
Neutral	100	90	1150	130
Mixed	50	110	30	1700

i. Calculate the model's accuracy. Comment if accuracy would be a good metric to reflect the model's performance here, given that the number of reviews in the class (Positive Feedback, Negative Feedback, Neutral Feedback, and Mixed Feedback) is 1000, 1200, 1500, and 2000, respectively. **[2]**

ii. Calculate each category's Macro Precision, Recall, and F1 Score. Then, comment on how the classification in the "Neutral Feedback" class can be improved, given its performance in the confusion matrix. **[2]**

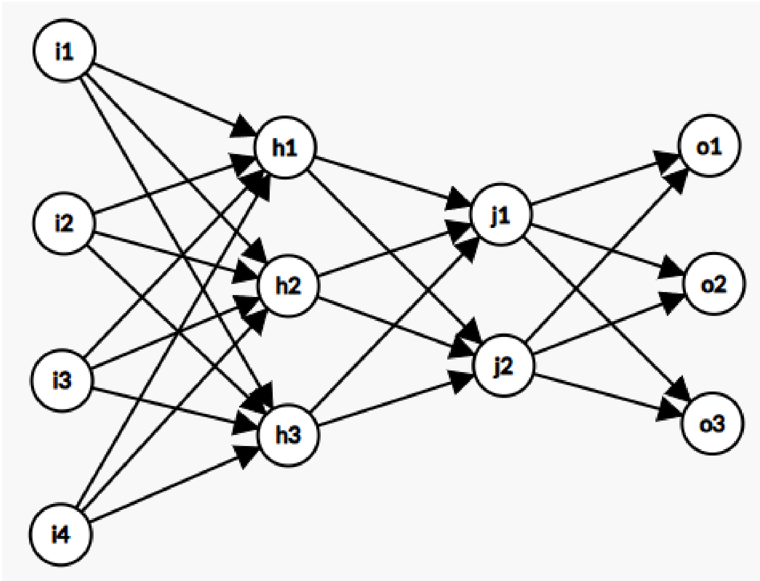
3. Consider the following pixel intensity values and their frequencies in an image:

- **145 - 10**
- **90 - 12**
- **220 - 15**
- **170 - 17**
- **180 - 21**
- **255 - 25**

Compressed Image size =
Codeword length + Frequency + Original Color length
= 200 + 100 + (6*8)
= 348 bits
Original Image size = 100 * 8 = 800 bits
Bits/pixel = 348 / 100 = 3.48 = 4

i. Use "Huffman Coding" to reduce the coding redundancy of this image. Start with frequency analysis and end at the Huffman table for this image **[2]**
ii. Then, find the image size after the compression. Also, find the compression ratio, relative data redundancy, and bits per pixel. **Consider that the original image used 8 bits per pixel.** **[3]**

4. Consider the following ANN:



Activation functions for the layers: **ReLU(layer 1), Sigmoid(layer 2), and suitable(layer 3/output layer) based on the output.**

inputs = **[0.25 0.3 0.45 0.15], outputs = [0.55 0.25 0.2]**

W_1 = [0.15 0.3 0.2], [0.25 0.6 0.1], [0.4 0.25 0.35] [0.6 0.4 0.15]

W_2 = [0.25 0.3], [0.35 0.2], [0.1 0.6]

W_{h2O} = [0.45 0.2 0.65] [0.3 0.5 0.45]

Now, for the given weights and input, determine the output of the ANN and the log loss using the given function: **[5]**

$$\text{Log Loss} = -y_i * \log(o_{outi}) - (1 - y_i) * \log(1 - o_{outi})$$

5. Consider the following **5x5** grayscale image: **[4]**

V = {0,1}

2	**1**	**0**	**0**	**1**
1	**2**	**1**	**0**	**0**
0	**1**	**0**	**1**	**1**
2	**1**	**0**	**1**	**0**
1	**0**	**1**	**1**	**0**

a. List all the connected components that are **4-adjacent**.
b. List all the connected components that are **8-adjacent.**
c. List all the connected components that are **m-adjacent.**